

EE2213 Introduction to AI

Lecture 8: Regression

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OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**

- What is AI
- Applications of AI
- AI agent

- **Search (Shaojing)**

- Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
- Informed search algorithms: greedy best-first, A*
- Applications

- **Optimisation (Shaojing)**

- Linear programming
- Convex problems
- Applications

- **Machine learning (Wang Si) (Weeks 6-9)**

- Supervised and unsupervised learning: regression, classification, clustering
- Neural networks and deep learning
- Applications

- **Knowledge representation (Wang Si) (Weeks 10-11)**

- Knowledge Representation and Reasoning
- Propositional Logic
- Applications

- **Ethical considerations (Shaojing)**

- Bias in AI
- Privacy concerns
- Societal impact

(Week 10, Monday)

**No Class on 20 Oct.! A recording
will be uploaded on Canvas.**

AGENDA

- We will discuss:
 - Linear Regression
 - Polynomial Regression
 - Ridge Regression (for regularization)
 - Evaluation Metrics

At the end of this lecture, you should be able to:

- ✓ Explain the working principles
- ✓ Apply the techniques to solve regression problems.
- ✓ Evaluate the performance of a regression task

Recap: Regression

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)$

Output: $f(\mathbf{x})$ that predict **numerical** y given $\mathbf{x}$

Applications:

House Price Prediction



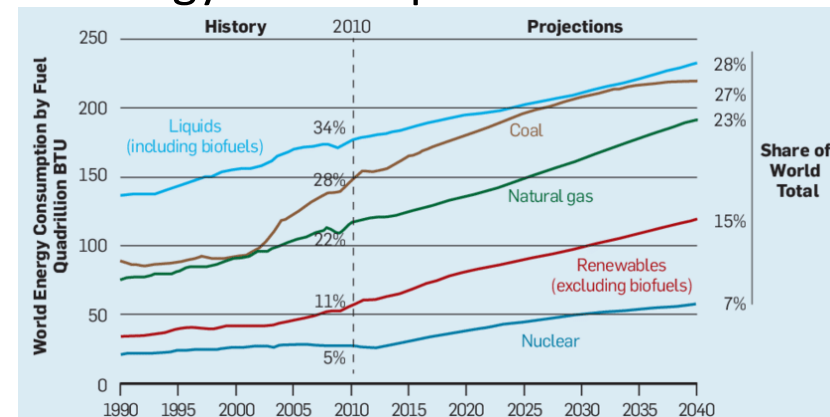
Student Performance Prediction

	Attendance (%)	Assignment	Quiz	Study Hours/Week	Final Exam
1	90	85	80	10	88
2	75	70	65	5	72
...	...	...	...	...	...

Sales Forecasting



Energy Consumption Estimation



Linear Regression

Linear Regression (Scalar Function)

- **Linear Regression** learns a model which is a **linear combination of features** of the input example.

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N),$

where $\mathbf{x}$ is a d -dimensional feature vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

Output: $f_{\mathbf{w}}(\mathbf{x}) = w_1x_1 + w_2x_2 + \dots + w_dx_d = \mathbf{x}^T \mathbf{w}$

where $\mathbf{w}$ is a d -dimensional vector of **parameters**

$$\mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_d \end{bmatrix}$$

e.g., House Price Prediction

No.	Size	No. rooms	Location	Sold Price
1	112	2	648310	600000
...				
90	100	4	129580	801256

Linear Regression (Scalar Function)

➤ Learning (Training)

Find the optimal value $\mathbf{w}^*$ which **minimizes**:

$$J(\mathbf{w}) = \sum_{i=1}^N \underbrace{(f_{\mathbf{w}}(\mathbf{x}_i) - y_i)^2}_{\text{Squared error loss}}$$

Least Squares Regression

- **Objective function**: the expression to be minimized or maximized
- **Loss function**: a measure of the difference between predicted output $f_{\mathbf{w}}(\mathbf{x}_i)$ and actual/desired/target output y_i .

$$\begin{aligned} & \sum_{i=1}^N (f_{\mathbf{w}}(\mathbf{x}_i) - y_i)^2 \\ &= (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) \end{aligned} \quad \text{where } \mathbf{X} = \begin{bmatrix} x_{1,1} & \dots & x_{1,d} \\ \vdots & \ddots & \vdots \\ x_{N,1} & \dots & x_{N,d} \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

Linear Regression (Scalar Function)

➤ Learning (Training)

$$\operatorname{argmin}_{\mathbf{w}} (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y})$$

$$\begin{aligned} J(\mathbf{w}) &= (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) = (\mathbf{w}^T \mathbf{X}^T - \mathbf{y}^T) (\mathbf{X}\mathbf{w} - \mathbf{y}) \\ &= (\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{w}^T \mathbf{X}^T \mathbf{y} - \mathbf{y}^T \mathbf{X} \mathbf{w} + \mathbf{y}^T \mathbf{y}) = (\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{y}^T \mathbf{X} \mathbf{w} + \mathbf{y}^T \mathbf{y}) \end{aligned}$$

The optimal value $\mathbf{w}^*$ should have **zero gradient**:

$$\frac{dJ(\mathbf{w})}{d\mathbf{w}} = \mathbf{0}$$

Why?

$$\frac{d(\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{y}^T \mathbf{X} \mathbf{w} + \mathbf{y}^T \mathbf{y})}{d\mathbf{w}} = \mathbf{0}$$

$$2\mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{X}^T \mathbf{y} = \mathbf{0}$$

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y}$$

If $\mathbf{X}^T \mathbf{X}$ is invertible, $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

$$\begin{aligned} \frac{d(\mathbf{w}^T \mathbf{A} \mathbf{w})}{d\mathbf{w}} &= (\mathbf{A} + \mathbf{A}^T) \mathbf{w} \\ \frac{d(\mathbf{y}^T \mathbf{A} \mathbf{w})}{d\mathbf{w}} &= \mathbf{A}^T \mathbf{y} \end{aligned}$$

Linear Regression (Scalar Function)

➤ Learning (Training)

- Geometric Interpretation of Least Squares Solution

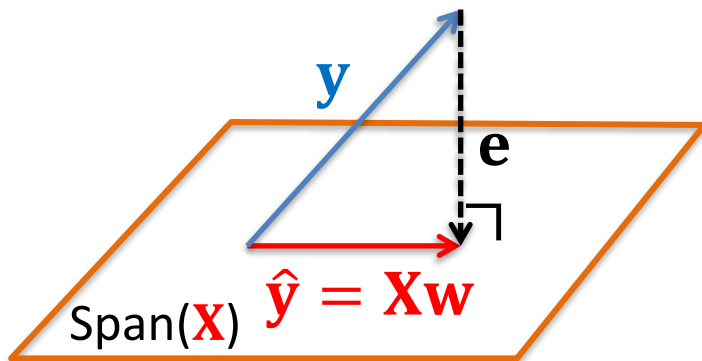
$$\underbrace{\mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y})}_{\mathbf{e}} = \mathbf{0}$$

$$\begin{bmatrix} x_{1,1} & \cdots & x_{N,1} \\ \vdots & \ddots & \vdots \\ x_{1,d} & \cdots & x_{N,d} \end{bmatrix} \mathbf{e} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$\mathbf{e}$ is perpendicular to each row of $\mathbf{X}^T$
(i.e., each column of $\mathbf{X}$)



$\mathbf{e}$ is perpendicular to the column space of $\mathbf{X}$
(i.e., $\text{Span}(\mathbf{X})$)



Linear Regression (Scalar Function)

➤ Prediction (Testing)

$$\mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_{new}) = \mathbf{X}_{new} \mathbf{w}^* = \begin{bmatrix} x_{1,1}^{new} & \dots & x_{1,d}^{new} \\ \vdots & \ddots & \vdots \\ x_{m,1}^{new} & \dots & x_{m,d}^{new} \end{bmatrix} \begin{bmatrix} w_1^* \\ \vdots \\ w_d^* \end{bmatrix}$$
$$= \begin{bmatrix} (\mathbf{x}_1^{new})^T \mathbf{w}^* \\ \vdots \\ (\mathbf{x}_m^{new})^T \mathbf{w}^* \end{bmatrix}$$

Is the obtained $\mathbf{w}^*$ a
global minimum?

Linear Regression (Scalar Function)

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6 = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using least squares regression.

Step 1: Learning (work out $\mathbf{w}^*$)

$$\mathbf{X} = \begin{bmatrix} -10 \\ -8 \\ -3 \\ -1 \\ 2 \\ 8 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Since $\mathbf{X}^T \mathbf{X} = [242]$ is invertible

$$= [242]^{-1} [-10, -8, -3, -1, 2, 8]$$

$$= -0.3512$$

$$\begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

Step 2: Prediction

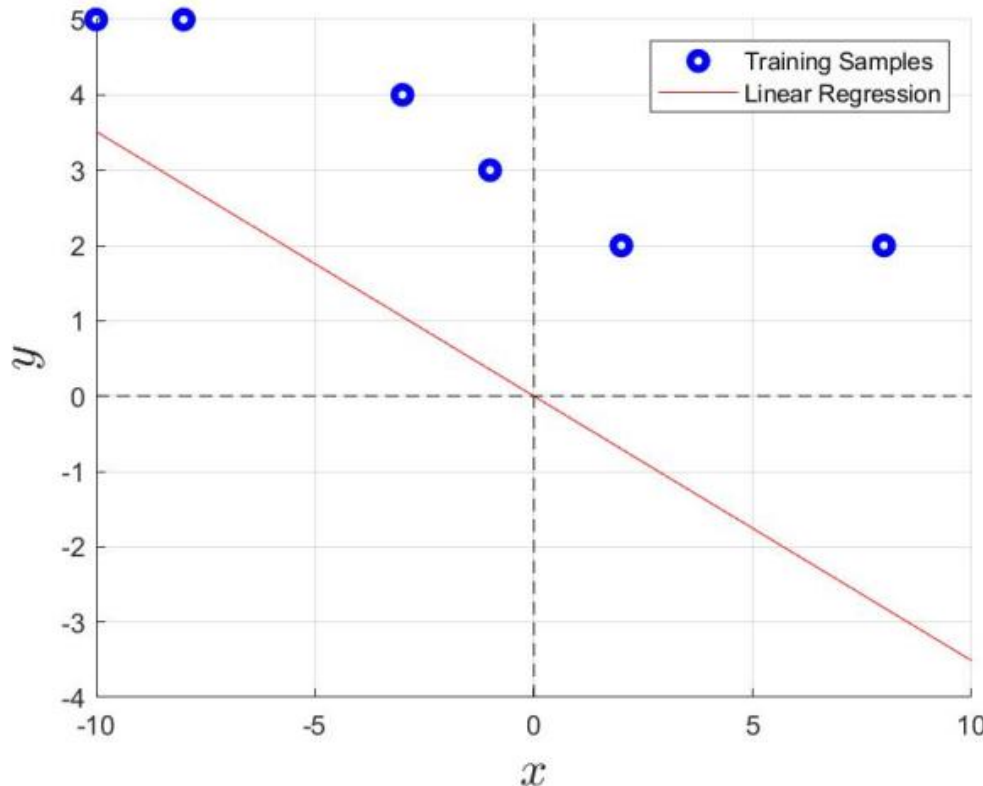
$$\mathbf{y}_t = \mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{x}_t \mathbf{w}^* = \begin{bmatrix} 7 \\ 9 \end{bmatrix} [-0.3512]$$

$$\mathbf{X}_t = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} -2.4587 \\ -3.1612 \end{bmatrix}$$

Linear Regression (Scalar Function)

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6 = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using least squares regression.



$$f_{\mathbf{w}}(\mathbf{x}) = w_1 x_1 = -0.3512 x_1$$

The line always passes through the origin.

Linear Regression (Scalar Function)

➤ Linear Regression with bias/offset

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N),$

Output: $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{x}^T \mathbf{w} = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_d x_d =$

where $\mathbf{x}$ is a $(d+1)$ -dimensional feature vector

$\mathbf{w}$ is a $(d+1)$ -dimensional vector of parameters

$$\mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ \vdots \\ w_d \end{bmatrix}$$

The bias/offset term w_0 is responsible for translating the line/plane/hyperplane away from the origin

Linear Regression (Scalar Function)

➤ Linear Regression with bias/offset

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N),$

Output: $f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \dots + w_dx_d = \mathbf{x}^T \mathbf{w}$

-Learning: $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

$$\mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ \vdots \\ w_d \end{bmatrix}$$

$$\text{where } \mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & \dots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \dots & x_{N,d} \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{X}_t \mathbf{w}^*$

$$\text{where } \mathbf{X}_t = \begin{bmatrix} 1 & x_{1,1}^t & \dots & x_{1,d}^t \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m,1}^t & \dots & x_{m,d}^t \end{bmatrix}, \mathbf{w}^* = \begin{bmatrix} w_0^* \\ \vdots \\ w_d^* \end{bmatrix}$$

Linear Regression (Scalar Function)

➤ Linear Regression with bias/offset

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6 = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using least squares regression.

Step 1: Learning (work out $\mathbf{w}^*$)

$$\mathbf{X} = \begin{bmatrix} 1 & -10 \\ 1 & -8 \\ 1 & -3 \\ 1 & -1 \\ 1 & 2 \\ 1 & 8 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Since $\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 6 & -12 \\ -12 & 242 \end{bmatrix}$ is invertible

$$= \begin{bmatrix} 6 & -12 \\ -12 & 242 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -10 \\ 1 & -8 \\ 1 & -3 \\ 1 & -1 \\ 1 & 2 \\ 1 & 8 \end{bmatrix}^T \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3.1055 \\ -0.1972 \end{bmatrix}$$

Step 2: Prediction

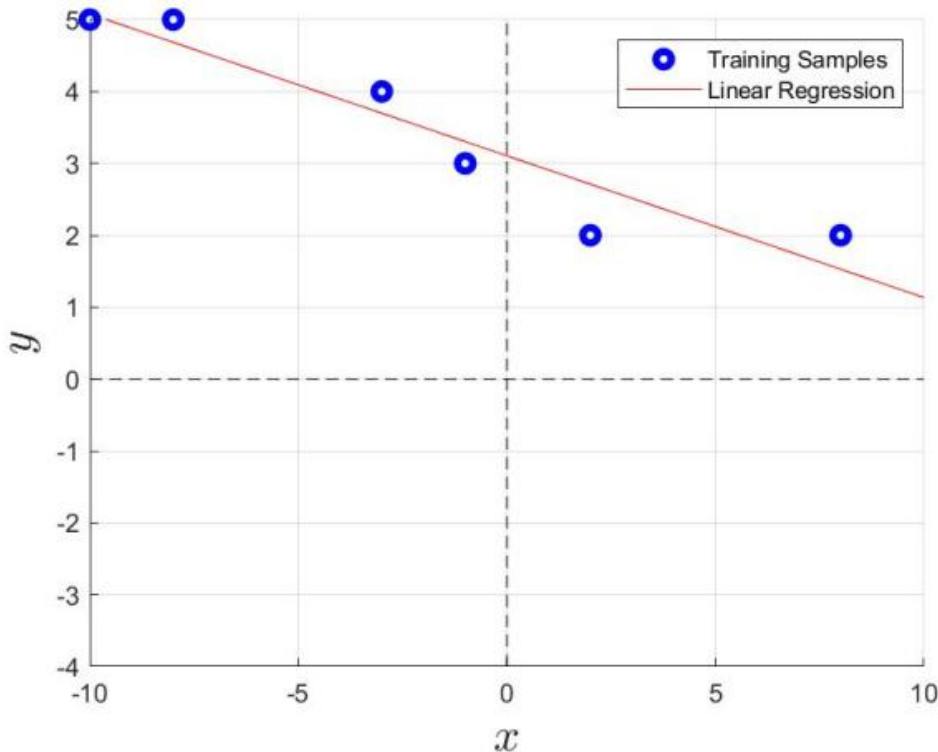
$$\mathbf{X}_t = \begin{bmatrix} 1 & 7 \\ 1 & 9 \end{bmatrix}$$

$$\begin{aligned} \mathbf{y}_t &= \mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{X}_t \mathbf{w}^* = \begin{bmatrix} 1 & 7 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} 3.1055 \\ -0.1972 \end{bmatrix} \\ &= \begin{bmatrix} 1.7248 \\ 1.3303 \end{bmatrix} \end{aligned}$$

Linear Regression (Scalar Function)

➤ Linear Regression with bias/offset

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6$
 $= \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test
dataset $\{7, 9\}$ using least squares regression.

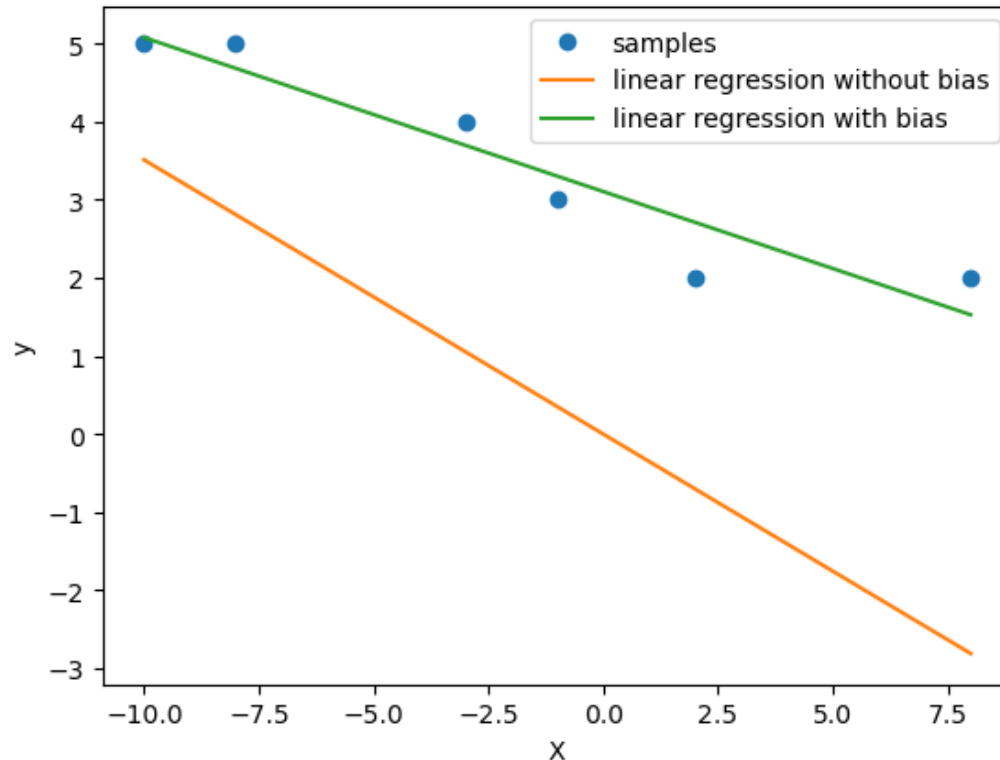


$$\begin{aligned} f_{\mathbf{w}}(\mathbf{x}) &= w_0 + w_1 x_1 \\ &= 1.7251 + 1.3307 x_1 \end{aligned}$$

Linear Regression (Scalar Function)

➤ Comparison

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6 = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using least squares regression.



Linear Regression (Vector Function)

➤ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_N, \mathbf{y}_N),$

where $\mathbf{x}$ is a $(d+1)$ -dimensional feature vector
 $\mathbf{y}$ is a h -dimensional target vector

$$\mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

e.g., predict multiple physiological signals

	Age	Weight	Physical activity level	Oxygen saturation	Heart rate	Blood pressure	Respiration rate
1							
...							

Output: $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \mathbf{x}^T \mathbf{W}$, where $\mathbf{W}$ is a matrix of size $(d+1) \times h$

$$\mathbf{W} = \begin{bmatrix} \boxed{w_{0,1}} & \cdots & \boxed{w_{0,h}} \\ \vdots & \ddots & \vdots \\ \boxed{w_{d,1}} & \cdots & \boxed{w_{d,h}} \end{bmatrix}$$

Equivalent to performing Linear Regression for each target output variable (h times)

$$\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = [f_{\mathbf{w}_1}(\mathbf{x}), \dots, f_{\mathbf{w}_h}(\mathbf{x})]$$

weight vector $\mathbf{w}_1$ for y_1

weight vector $\mathbf{w}_h$ for y_h

Linear Regression (Vector Function)

➤ Learning

Find the optimal value $\mathbf{W}^*$ which minimizes:

$$J(\mathbf{W}) = \sum_{k=1}^h \sum_{i=1}^N (f_{\mathbf{w}_k}(\mathbf{x}_i) - y_{i,k})^2$$

$$= \sum_{k=1}^h (\mathbf{X}\mathbf{w}_k - \mathbf{y}_k)^T (\mathbf{X}\mathbf{w}_k - \mathbf{y}_k)$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & \dots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \dots & x_{N,d} \end{bmatrix}$$

The optimal value $\mathbf{W}^*$ should have **zero gradient**:

$$\frac{dJ(\mathbf{W})}{d\mathbf{W}} = \begin{bmatrix} \frac{\partial J}{\partial w_{0,1}} \\ \vdots \\ \frac{\partial J}{\partial w_{d,1}} \end{bmatrix} \dots \begin{bmatrix} \frac{\partial J}{\partial w_{0,h}} \\ \vdots \\ \frac{\partial J}{\partial w_{d,h}} \end{bmatrix} = \mathbf{0}$$

➔

$$\frac{dJ(\mathbf{W})}{d\mathbf{w}_k} = \mathbf{0}, k = 1, 2, \dots, h$$

⬇

$$\mathbf{w}_k^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_k$$

Linear Regression (Vector Function)

➤ Learning

Find the optimal value for $\mathbf{W}^*$ which minimizes:

$$\begin{aligned} J(\mathbf{W}) &= \sum_{k=1}^h \sum_{i=1}^N (f_{\mathbf{w}_k}(\mathbf{x}_i) - y_{i,k})^2 \\ &= \sum_{k=1}^h (\mathbf{X}\mathbf{w}_k - \mathbf{y}_k)^T (\mathbf{X}\mathbf{w}_k - \mathbf{y}_k) \end{aligned}$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \cdots & x_{N,d} \end{bmatrix}$$

$$\begin{aligned} \mathbf{W}^* &= [\mathbf{w}_1^*, \dots, \mathbf{w}_h^*] = [(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_1, \dots, (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_h] \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \end{aligned}$$

$$\mathbf{Y} = \begin{bmatrix} y_{1,1} & \cdots & y_{1,h} \\ \vdots & \ddots & \vdots \\ y_{N,1} & \cdots & y_{N,h} \end{bmatrix}$$

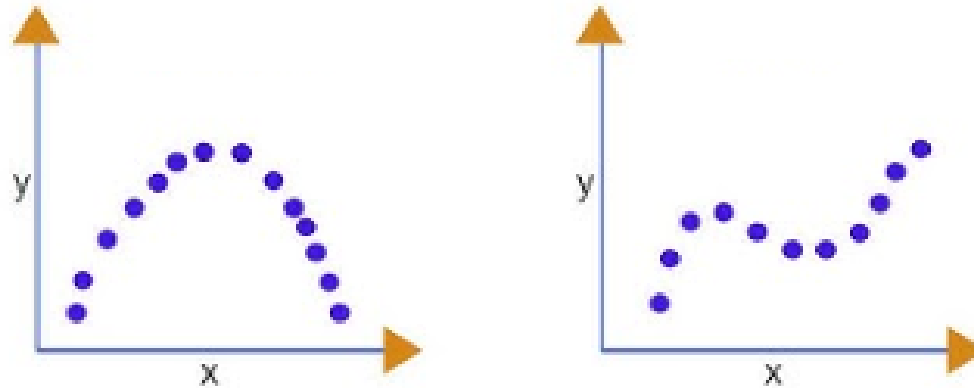
Is the obtained $\mathbf{W}^*$ a
global minimum?

Linear Regression (Vector Function)

➤ Prediction

$$\mathbf{F}_{\mathbf{W}^*}(\mathbf{X}_t) = \mathbf{X}_t \mathbf{W}^* = \begin{bmatrix} \mathbf{1} x_{1,1}^t & \cdots & x_{1,d}^t \\ \vdots & \ddots & \vdots \\ \mathbf{1} x_{m,1}^t & \cdots & x_{m,d}^t \end{bmatrix} \begin{bmatrix} w_{0,1}^* & \cdots & w_{0,h}^* \\ \vdots & \ddots & \vdots \\ w_{d,1}^* & \cdots & w_{d,h}^* \end{bmatrix}$$

$$= \begin{bmatrix} (\mathbf{x}_1^t)^T \mathbf{w}_1^* & \cdots & (\mathbf{x}_1^t)^T \mathbf{w}_h^* \\ \vdots & \ddots & \vdots \\ (\mathbf{x}_m^t)^T \mathbf{w}_1^* & \cdots & (\mathbf{x}_m^t)^T \mathbf{w}_h^* \end{bmatrix}$$



Polynomial Regression

non-linear relationship between input and output

Polynomial Regression

➤ Polynomial Model

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + \sum_{i=1}^d w_i x_i + \sum_{i=1}^d \sum_{j=1}^d w_{ij} x_i x_j + \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d w_{ijk} x_i x_j x_k + \dots$$

1st order (Linear)

2nd order (Quadratic)

3rd order (Cubic)

Handwritten: $w_0 + w_1 x_1 + w_{11} x_1^2$
Order number:
Hyperparameter!

e.g., $d=1$, 3rd order: $f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_{11} x_1^2 + w_{111} x_1^3$

$d=2$, 2nd order: $f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2 + w_{11} x_1^2 + w_{12} x_1 x_2 + w_{22} x_2^2$

No. parameters for n -th order polynomial model with d input variables?

$$\binom{n+d}{d} = \frac{(n+d)!}{n! d!}$$

No. parameters for *degree n terms* with d input variables?

$$\binom{n+d-1}{d-1} = \frac{(n+d-1)!}{n! (d-1)!}$$

Handwritten: $w_2 x_1 x_2$
 $w_{11} x_1^2$
 $w_{22} x_2^2$

- Proof: total No. paras for n -th order polynomial model with d input variables is $\binom{n+d}{d} = \frac{(n+d)!}{n!d!}$

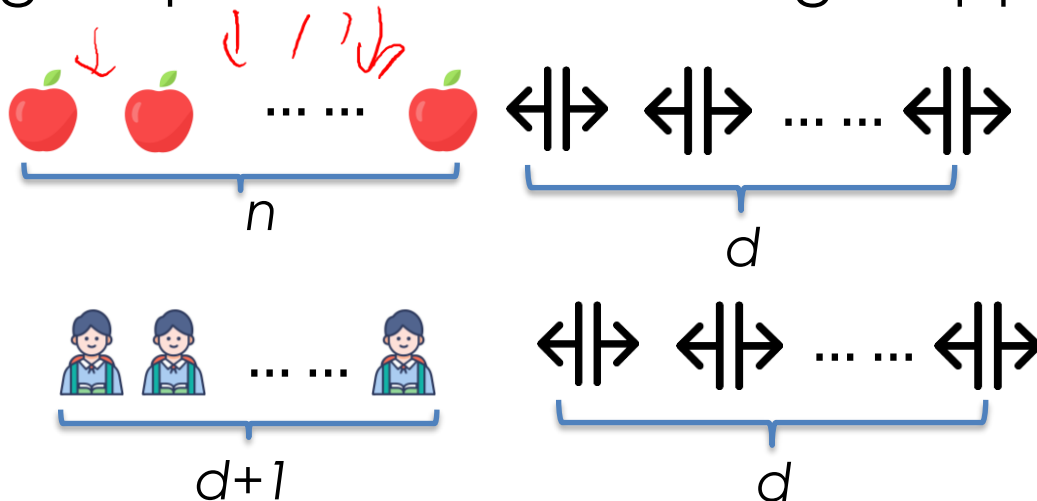
$$x_1^{k_1} x_2^{k_2} \dots x_d^{k_d} : k_1 + k_2 + \dots + k_d \leq n$$



$$k_1 + k_2 + \dots + k_d + k_{d+1} = n$$

Counting: equivalent to allocating n apples to $d+1$ students.

$\binom{n+d}{d}$

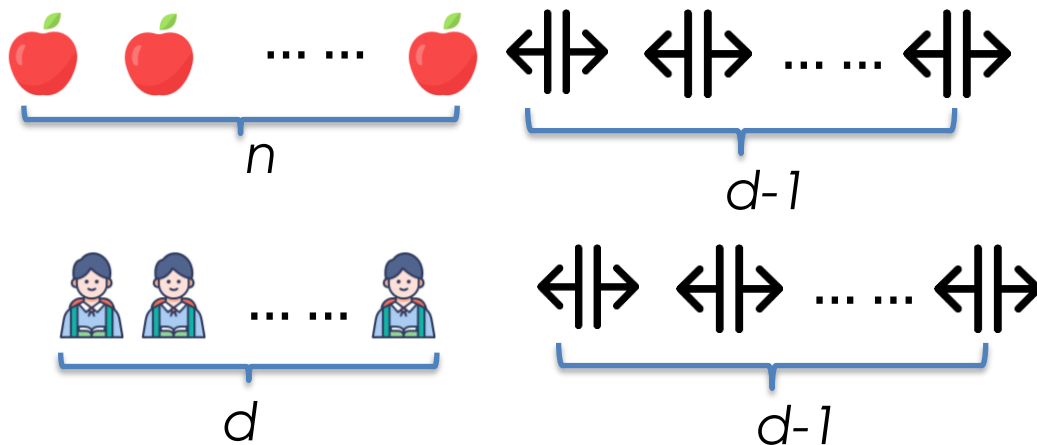


$$\binom{n+d}{d} = \binom{n+d}{n}$$

- Proof: No. paras for degree n terms with d input variables is $\binom{n+d-1}{d-1} = \frac{(n+d-1)!}{n!(d-1)!}$

$$x_1^{k_1} x_2^{k_2} \dots x_d^{k_d} : k_1 + k_2 + \dots + k_d = n$$

Counting: equivalent to allocating n apples to d students.



$$\binom{n+d-1}{d-1} = \binom{n+d-1}{n}$$

Polynomial Regression

➤ Polynomial Model

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + \sum_{i=1}^d w_i x_i + \sum_{i=1}^d \sum_{j=1}^d w_{ij} x_i x_j + \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d w_{ijk} x_i x_j x_k + \dots$$

$$= f_{\mathbf{w}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T \mathbf{w}$$

$$\mathbf{p} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \\ \vdots \\ x_i x_j \\ \vdots \\ x_i x_j x_k \\ \vdots \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \\ \vdots \\ w_{ij} \\ \vdots \\ w_{ijk} \\ \vdots \end{bmatrix}$$

Equivalent to:

polynomial transformation
+
linear regression

Polynomial Regression

➤ Learning

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N),$

Output: $f_{\mathbf{w}}(\mathbf{x}) = f_{\mathbf{w}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T \mathbf{w}$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Find the optimal value $\mathbf{w}^*$ which minimizes:

$$J(\mathbf{w}) = \sum_{i=1}^N (f_{\mathbf{w}}(\mathbf{p}(\mathbf{x}_i)) - y_i)^2 = (\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y})$$

$$\mathbf{P} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,d} & \cdots & x_{1,i}x_{1,j} & \cdots & x_{1,i}x_{1,j}x_{1,k} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{N,1} & \cdots & x_{N,d} & \cdots & x_{N,i}x_{N,j} & \cdots & x_{N,i}x_{N,j}x_{N,k} & \cdots \end{bmatrix}$$

Solution: $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$, If $\mathbf{P}^T \mathbf{P}$ is invertible

global
minimum?

Polynomial Regression

➤ Prediction

$$\mathbf{f}_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{w}^*$$

$$= \begin{bmatrix} 1 & x_{1,1}^t & \cdots & x_{1,d}^t & \cdots & x_{1,i}^t x_{1,j}^t & \cdots & x_{1,i}^t x_{1,j}^t x_{1,k}^t & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{m,1}^t & \cdots & x_{m,d}^t & \cdots & x_{m,i}^t x_{m,j}^t & \cdots & x_{m,i}^t x_{m,j}^t x_{m,k}^t & \cdots \end{bmatrix} \begin{bmatrix} w_0^* \\ w_1^* \\ \vdots \\ w_d^* \\ \vdots \\ w_{ij}^* \\ \vdots \\ w_{ijk}^* \\ \vdots \end{bmatrix}$$

$$= \begin{bmatrix} (\mathbf{p}_1^t)^T \mathbf{w}^* \\ \vdots \\ (\mathbf{p}_m^t)^T \mathbf{w}^* \end{bmatrix}$$

Polynomial Regression

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6 = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using 3rd order polynomial regression.

Step 0: Polynomial Transformation ($\mathbf{X} \rightarrow \mathbf{P}$)

3rd order polynomial model with 1 input variable:

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_{11} x_1^2 + w_{111} x_1^3 = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_{11} \\ w_{111} \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} 1 & x_{1,1} & x_{1,1}^2 & x_{1,1}^3 \\ 1 & x_{2,1} & x_{2,1}^2 & x_{2,1}^3 \\ 1 & x_{3,1} & x_{3,1}^2 & x_{3,1}^3 \\ 1 & x_{4,1} & x_{3,1}^2 & x_{4,1}^3 \\ 1 & x_{5,1} & x_{4,1}^2 & x_{5,1}^3 \\ 1 & x_{6,1} & x_{4,1}^2 & x_{5,1}^3 \\ & & x_{6,1}^2 & x_{6,1}^3 \end{bmatrix} = \begin{bmatrix} 1 & -10 & 100 & -1000 \\ 1 & -8 & 64 & -512 \\ 1 & -3 & 9 & -27 \\ 1 & -1 & 1 & -1 \\ 1 & 2 & 4 & 8 \\ 1 & 8 & 64 & 512 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

Polynomial Regression

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6 = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using 3rd order polynomial regression.

Step 1: Learning (work out $\mathbf{w}^*$)

$$\text{Since } \mathbf{P}^T \mathbf{P} \text{ is invertible, } \mathbf{w}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y} = \begin{bmatrix} 2.6894 \\ -0.3772 \\ 0.0134 \\ 0.0029 \end{bmatrix}$$

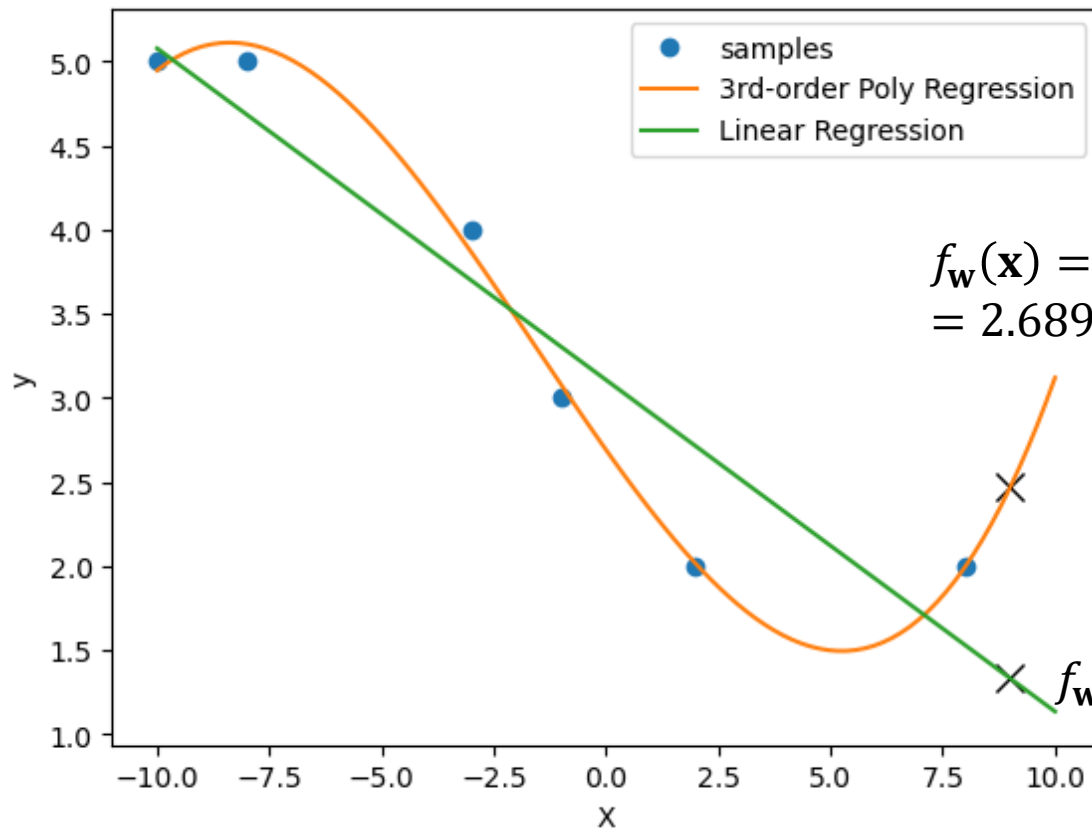
Step 2: Prediction

$$\mathbf{P}_t = \begin{bmatrix} 1 & x_{1,1}^t & (x_{1,1}^t)^2 & (x_{1,1}^t)^3 \\ 1 & x_{2,1}^t & (x_{2,1}^t)^2 & (x_{2,1}^t)^3 \end{bmatrix} = \begin{bmatrix} 1 & 7 & 49 & 343 \\ 1 & 9 & 81 & 729 \end{bmatrix}$$

$$\begin{aligned} \mathbf{y}_t &= \mathbf{f}_{\mathbf{w}^*}(\mathbf{P}_t) = \mathbf{P}_t \mathbf{w}^* = \begin{bmatrix} 1 & 7 & 49 & 343 \\ 1 & 9 & 81 & 729 \end{bmatrix} \begin{bmatrix} 2.6894 \\ -0.3772 \\ 0.0134 \\ 0.0029 \end{bmatrix} \\ &= \begin{bmatrix} 1.6874 \\ 2.4661 \end{bmatrix} \end{aligned}$$

Polynomial Regression

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^6 = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using 3rd order polynomial regression.



$$\begin{aligned} f_{\mathbf{w}}(\mathbf{x}) &= w_0 + w_1 x_1 + w_{11} x_1^2 + w_{111} x_1^3 \\ &= 2.6894 - 0.3772 x_1 + 0.0134 x_1^2 + 0.0029 x_1^3 \end{aligned}$$

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 = 1.7251 + 1.3307 x_1$$

Polynomial Regression

➤ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_N, \mathbf{y}_N),$

where $\mathbf{x}$ is a d -dimensional feature vector

$\mathbf{y}$ is a h -dimensional target vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

Output: $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \mathbf{f}_{\mathbf{W}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T \mathbf{W},$

-Learning: $\mathbf{W}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{W}^*$

Ridge Regression

If $\mathbf{X}^T \mathbf{X}$ or $\mathbf{P}^T \mathbf{P}$ is not invertible

Ridge Regression

➤ Ridge Regression = Regression + L2 regularization

Shrinks the regression coefficients w by imposing a penalty to their size.

Find the optimal value $\mathbf{w}^*$ which minimizes:

$$J(\mathbf{w}) = \sum_{i=1}^N (f_{\mathbf{w}}(\mathbf{p}(\mathbf{x}_i)) - y_i)^2 + \lambda \sum_i w_i^2$$

L2 regularization:
 $\sum_i w_i^2 = \|\mathbf{w}\|_2^2$,
squared L2 norm of $\mathbf{w}$

$$= (\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w}$$

(Hyperparameter)

$\lambda > 0$ is a complexity/regularization parameter that controls the amount of shrinkage: the larger the value of λ , the greater the amount of shrinkage.

Ridge Regression

➤ Learning

$$\underset{\mathbf{w}}{\operatorname{argmin}} (\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w}$$

Solution: $\frac{dJ(\mathbf{w})}{d\mathbf{w}} = \mathbf{0}$

$$\frac{d \left((\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \right)}{d\mathbf{w}} = \mathbf{0}$$

$$2\mathbf{P}^T \mathbf{P}\mathbf{w} - 2\mathbf{P}^T \mathbf{y} + 2\lambda \mathbf{w} = \mathbf{0}$$

$$\mathbf{P}^T \mathbf{P}\mathbf{w} + \lambda \mathbf{w} = \mathbf{P}^T \mathbf{y}$$

$$(\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I}) \mathbf{w} = \mathbf{P}^T \mathbf{y}$$

Since $\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I}$ is always invertible, $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I})^{-1} \mathbf{P}^T \mathbf{y}$

Ridge Regression

Proof: $\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I}$ is always invertible when $\lambda > 0$

Suppose $\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I}$ is not invertible.



Columns/rows in $\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I}$ are linearly dependent



There exists a vector $\mathbf{a} \neq \mathbf{0}$, s.t., $(\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I})\mathbf{a} = \mathbf{0}$



$$\mathbf{a}^T (\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I}) \mathbf{a} = 0$$



$$\mathbf{a}^T \mathbf{P}^T \mathbf{P} \mathbf{a} + \lambda \mathbf{a}^T \mathbf{a} = 0$$



$$\|\mathbf{P}\mathbf{a}\|^2 + \lambda \|\mathbf{a}\|^2 = 0$$

However, $\|\mathbf{P}\mathbf{a}\|^2 + \lambda \|\mathbf{a}\|^2 = 0$ only holds if $\|\mathbf{P}\mathbf{a}\|^2 = \|\mathbf{a}\|^2 = 0$



$$\mathbf{a} = \mathbf{0}$$



Columns/rows in $\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I}$ are linearly independent

Ridge Regression

➤ Learning

Geometric Interpretation of Ridge Regression Solution

$$\underset{\mathbf{w}}{\operatorname{argmin}} (\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \quad \boxed{\text{Dual form}}$$

$\mathbf{w}$
**Lagrangian
Duality**

$$\underset{x}{\operatorname{argmin}} f(x) + \lambda g(x)$$

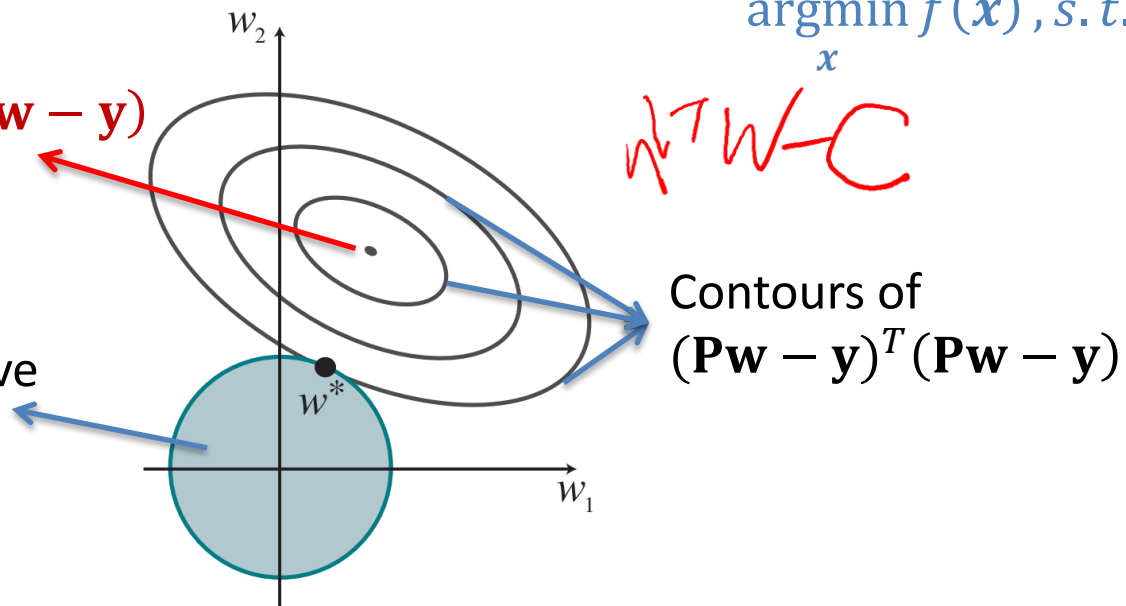
$$\underset{\mathbf{w}}{\operatorname{argmin}} (\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y}), \text{ subject to } \mathbf{w}^T \mathbf{w} \leq c \quad \boxed{\text{Primal form}}$$

$$\underset{x}{\operatorname{argmin}} f(x), s.t. g(x) \leq 0$$

$$\underset{\mathbf{w}}{\operatorname{argmin}} (\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y})$$

$\mathbf{w}^T \mathbf{w} - c$

Set of points $\mathbf{w}$ in 2D
weight space that have
 $\mathbf{w}^T \mathbf{w} \leq c$



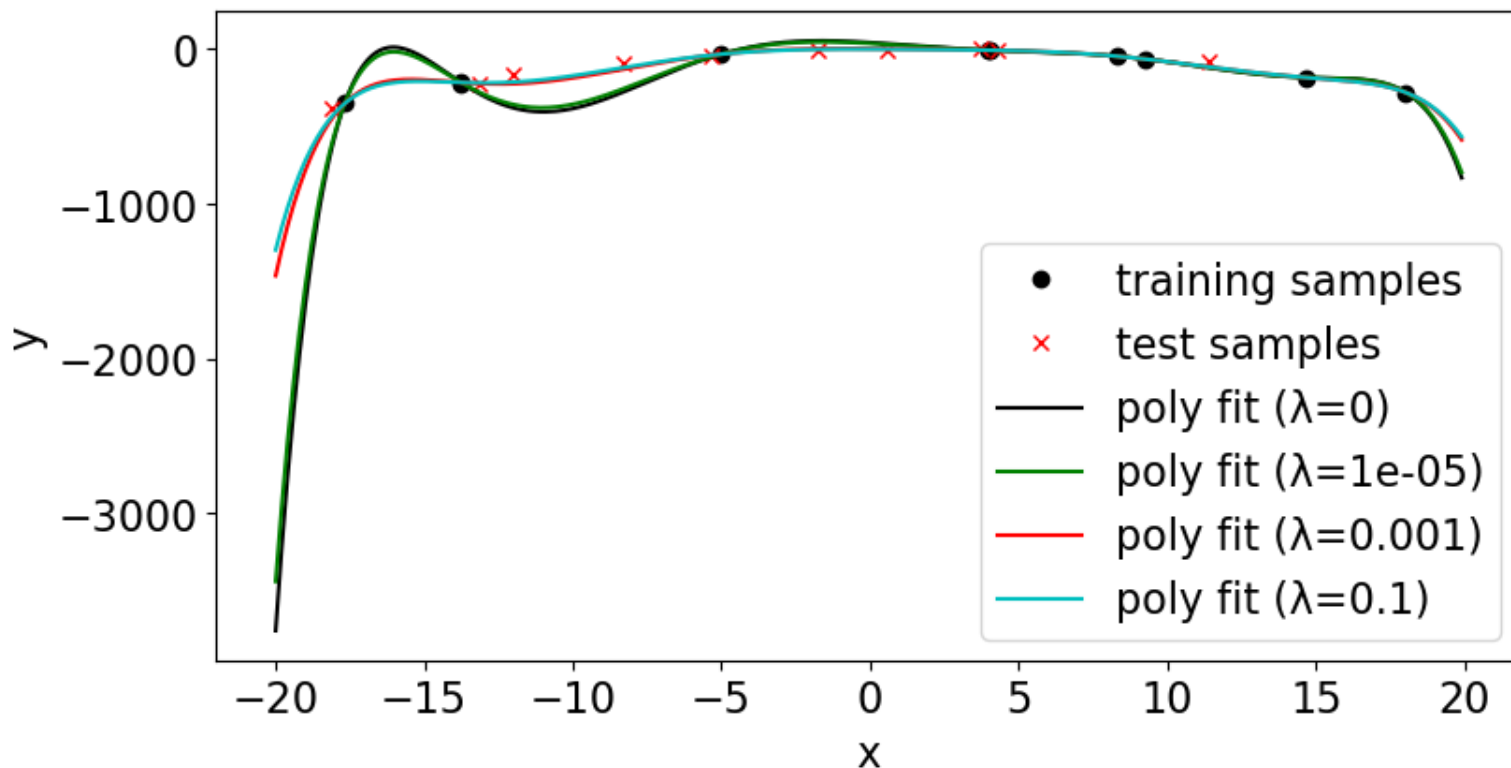
Ridge Regression

➤ Prediction

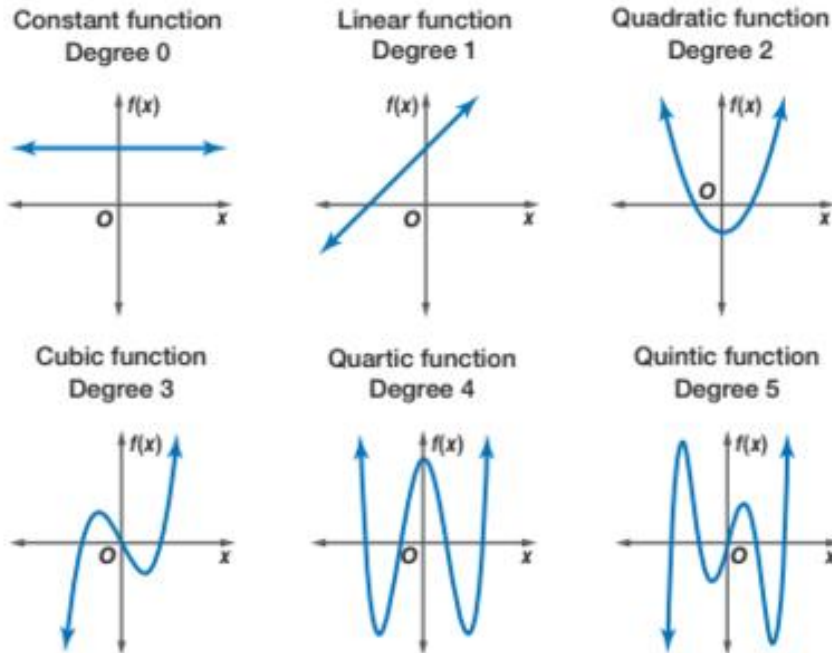
$$\mathbf{f}_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{w}^*$$

How is weight shrinkage connected with $\mathbf{P}^T \mathbf{P}$ being not invertible?

Example: Order = 8, no. of input variable = 1



Polynomial model with different degrees



Flattening
(Fewer ups and downs)



Reducing degree
(Reduce no. polynomial features)

When is $\mathbf{P}^T \mathbf{P}$ not invertible? When $\text{rank}(\mathbf{P}) < \text{no. polynomial features}$

$$\text{Rank}(\mathbf{P}) = \text{Rank}(\mathbf{P}^T \mathbf{P})$$



Can be proved by **rank-nullity theorem**

$$(\text{Rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = \text{No. cols})$$

Intrinsic dimension of the
feature space

Prove $\text{Rank}(\mathbf{P}) = \text{Rank}(\mathbf{P}^T \mathbf{P})$ by rank-nullity theorem



Prove $\text{Nullity}(\mathbf{P}) = \text{Nullity}(\mathbf{P}^T \mathbf{P})$



Prove { if $\mathbf{x}$ satisfies $\mathbf{P}\mathbf{x} = \mathbf{0}$, $\mathbf{P}^T \mathbf{P}\mathbf{x} = \mathbf{0}$
 and
 if $\mathbf{x}$ satisfies $\mathbf{P}^T \mathbf{P}\mathbf{x} = \mathbf{0}$, $\mathbf{P}\mathbf{x} = \mathbf{0}$

$$\mathbf{x}^T \mathbf{P}^T \mathbf{P} \mathbf{x} = 0$$



$$\|\mathbf{P}\mathbf{x}\|^2 = 0$$



$$\mathbf{P}\mathbf{x} = \mathbf{0}$$

Ridge Regression

➤ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_N, \mathbf{y}_N),$

where $\mathbf{x}$ is a d -dimensional feature vector
 $\mathbf{y}$ is a h -dimensional target vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

Output: $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \mathbf{f}_{\mathbf{W}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T \mathbf{W},$

-Learning:

Find the optimal values for $\mathbf{W}^*$ which minimizes:

$$\begin{aligned} J(\mathbf{W}) &= \sum_{k=1}^h \left(\sum_{i=1}^N (f_{\mathbf{w}_k}(\mathbf{p}(\mathbf{x}_i)) - y_{i,k})^2 + \lambda \sum_i w_{i,k}^2 \right) \\ &= \sum_{k=1}^h ((\mathbf{P}\mathbf{w}_k - \mathbf{y}_k)^T (\mathbf{P}\mathbf{w}_k - \mathbf{y}_k) + \lambda \mathbf{w}_k^T \mathbf{w}_k) \end{aligned}$$

$$\mathbf{Y} = \begin{bmatrix} y_{1,1} & \cdots & y_{1,h} \\ \vdots & \ddots & \vdots \\ y_{N,1} & \cdots & y_{N,h} \end{bmatrix}$$

Solution in Compact Form: $\mathbf{W}^* = (\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I})^{-1} \mathbf{P}^T \mathbf{Y}$

Ridge Regression

➤ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_N, \mathbf{y}_N),$

where $\mathbf{x}$ is a d -dimensional feature vector
 $\mathbf{y}$ is a h -dimensional target vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

Output: $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \mathbf{f}_{\mathbf{W}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T \mathbf{W},$

-Learning: $\mathbf{W}^* = (\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I})^{-1} \mathbf{P}^T \mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{W}^*$

Evaluation Metrics for Regression

Evaluation Metrics for Regression

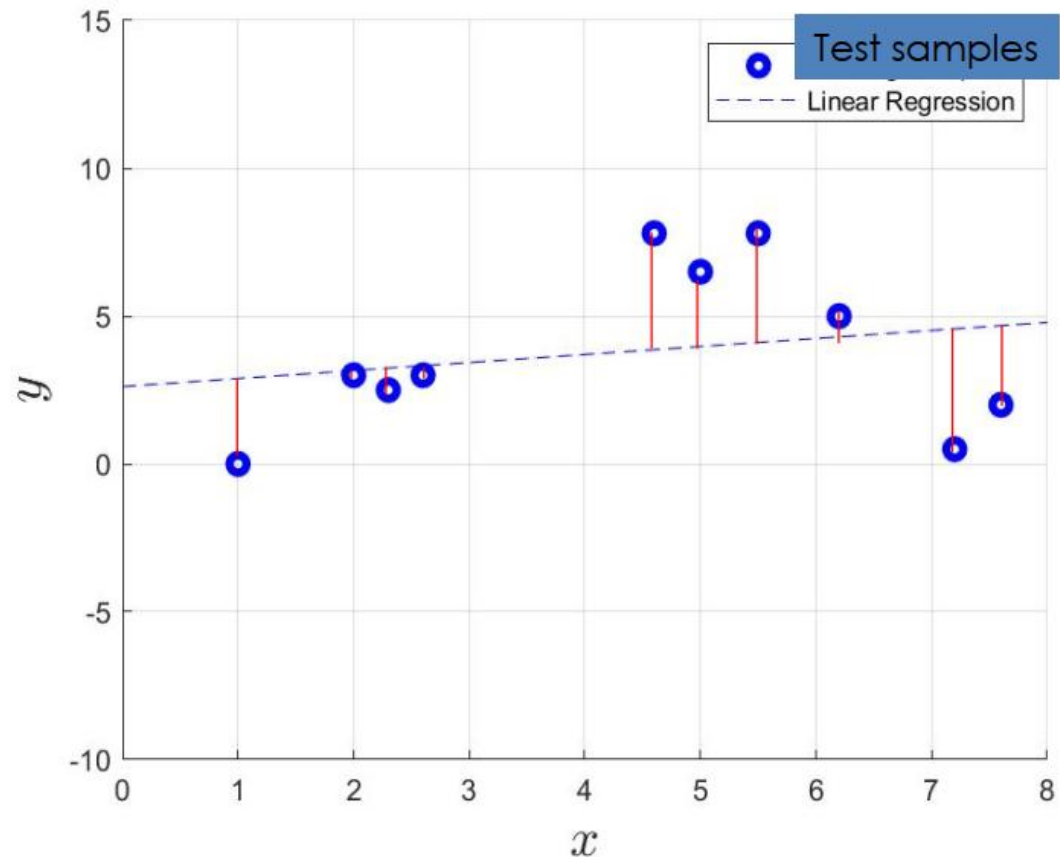
Mean Square Error

$$(\text{MSE} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n})$$

Mean Absolute Error

$$(\text{MAE} = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n})$$

where y_i denotes the target output and $\hat{y}_i$ denotes the predicted output for sample i .



```
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

```
mae = mean_absolute_error(y_true, y_pred)
```

```
mse = mean_squared_error(y_true, y_pred)
```

Summary

✓ Linear Regression

Scalar output for each sample

-Learning: $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{X}_t \mathbf{w}^*$

Vector output for each sample

-Learning: $\mathbf{W}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{X}_t) = \mathbf{X}_t \mathbf{W}^*$

✓ Polynomial Regression

Scalar output for each sample

-Learning: $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{w}^*$

Vector output for each sample

-Learning: $\mathbf{W}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{W}^*$

✓ Ridge Regression

Scalar output for each sample

-Learning: $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I})^{-1} \mathbf{P}^T \mathbf{y}$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{w}^*$

Vector output for each sample

-Learning: $\mathbf{W}^* = (\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I})^{-1} \mathbf{P}^T \mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t \mathbf{W}^*$

✓ 2 Evaluation Metrics for Regression Task

THE END